MTH401 - Differential Equations (Quiz-4) **Quiz Start Time: 02:21 PM, 08 January 2026**

Question # 6 of 10 (Start time: 02:22:34 PM, 08 January 2026) **Total Marks:** 1

The simultaneous ordinary differential equations involve two or more equations that contain derivatives of two or more unknown functions of _____ independent variable.

Select the correct option

Reload Math Equations

- ○ infinite
- ○ 2
- ◉ 1
- ○ 2

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz-4)

Question # 5 of 10 (Start time: 02:22:22 PM, 08 January 2026)

Spherical Bessel functions are used in many physical problems.

Select the correct option

- (●) True
- ( ) False

MTH401 - Differential Equations (Quiz-4)

Question # 4 of 10 (Start time: 02:21:53 PM, 08 January 2026)

The linear normal form of

$$2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 5y = 0$$

,by using

$$y = x_1, y' = x_1' \text{and } y'' = x_2'$$

, is_____.

Select the correct option

○	$x_1' = x_2, x_2' = 2x_2 + \frac{5}{2}x_1$
●	$x_1' = x_2, x_2' = -2x_2 - \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = 2x_2 - \frac{5}{2}x_1$
○	$x_1' = x_2, x_2' = -2x_2 + \frac{5}{2}x_1$

MTH401 - Differential Equations (Quiz-4)

Question # 3 of 10 (Start time: 02:21:40 PM, 08 January 2026)

A function f is said to be analytic at a point a if it can be represented by a power series in (x-a) with a positive radius of_____.

Select the correct option

◉	convergence
○	divergence

MTH401 - Differential Equations (Quiz-4)

Question # 2 of 10 (Start time: 02:21:21 PM, 08 January 2026)

The equation, $(x^2+1)\frac{d^2y}{dx^2}+2x\frac{dy}{dx}+6y=0$, has the singular point(s) at $x=$ _____.

Select the correct option

○	$(I)\quad \pm 1$
○	$(II)\quad i$
◉	$(IV)\quad$ Both (II) and (III).
○	$(III)\quad -i$

Click

MTH401 - Differential Equations (Quiz-4) Quiz

Question # 1 of 10 (Start time: 02:21:04 PM, 08 January 2026)

The equation, $(x^2-4)^2\frac{d^2y}{dx^2}+(x-2)\frac{dy}{dx}+y=0$, has the singular point(s) at $x=$ _____.

Select the correct option

◉	± 2
○	0
○	± 1
○	$\pm i$

Click to

MTH401 - Differential Equations (Quiz-4)

Question # 10 of 10 (Start time: 02:23:39 PM, 08 January 2026)

A second order linear differential equation of the form

$$(1-x^2)y - 2xy' + n(n+1)y = 0$$

is called _____ differential equation.

Select the correct option

○	Bessel
○	Bernoulli
○	Picard
◉	Legendre

Question # 9 of 10 (Start time: 02:23:18 PM, 08 January 2026)

Every n^{th} order differential equation, $y^{(n)} = F\left(t, y, y', \ldots, y^{(n-1)}\right)$, as well as most systems of differential equations, could be reduced to the _____ order system.

Select the correct option

- $n+1$
- infinite
- $n-1$
- n^{th}

MTH401 - Differential Equations (Quiz-4)

Question # 8 of 10 (Start time: 02:23:05 PM, 08 January 2026)

The method of Frobenius always provide two solutions.

Select the correct option

- ○ True
- ◉ False

MTH401 - Differential Equations (Quiz-4)

Question # 7 of 10 (Start time: 02:22:47 PM, 08 January 2026)

Bessel functions of index half an _____ integer are called Spherical Bessel functions.

Select the correct option

- (●) odd
- ( ) even